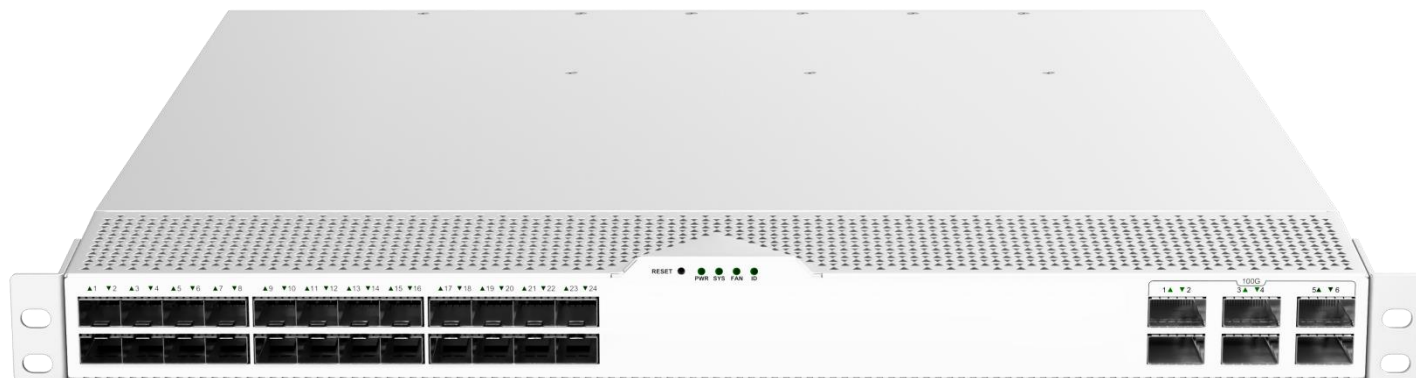


# Data Center White Box Switch

## S6750-24X6CQM-AC



S6750-24X6CQM-AC is high performance, multi service aggregation or ToR (Top of Rack) switch for telco access, security and cloud, with 24 x 10G (SFP+) + 6 x 100G/40G (QSFP28) interfaces.

S6750-24X6CQM-AC switch is an advanced best-in-class, open-source network platform that line-rate L2/ L3 up to 4 Tbps switching performance of Marvell Prestera is paired with multi service feature. It is well suited for enterprise access/aggregation, high bandwidth interconnection for video distribution and security. It can also be used as the cost effective ToR switch for private cloud. S6750-24X6CQM-AC is Compact 1RU standard based open architecture, both its power supply and fan modules are hot-swappable, supporting 1+1 and 4+1 redundancy respectively.

The S6750-24X6CQM-AC switch can be deployed as a top- of- rack (ToR) or leaf switch supporting 10GbE to servers with 100GbE uplinks, each 100G QSFP28 port can be configured 40G as well as 4 x 25GbE or 4 x 10GbE via breakout cables.

### Hardware Specification

|                  |                                                                         |
|------------------|-------------------------------------------------------------------------|
| Model            | S6750-24X6CQM-AC                                                        |
| Business Ports   | 24*10G SFP+ ports<br>6*40/100G QSFP28 ports                             |
| Management ports | 1x RJ45 100/1000Base-T out-of-band<br>1x RJ-45 Console<br>1x USB Type A |
| Switch ASIC      | Marvell Aldrin 98DX8548                                                 |
| CPU              | Maverll Octeon TX2 CN9130                                               |

|                           |                                                                   |
|---------------------------|-------------------------------------------------------------------|
| Memory                    | 8GB DDR4                                                          |
| Storage                   | 8GB eMMC NAND FLASH<br>SSD module optional (default 64GB)         |
| Power supply units        | PSU Input: 110V–240 VAC,<br>PSU Output: 350W<br>1+1 hot pluggable |
| Power consumption         | 250W max                                                          |
| Fan unit                  | 5 (4+1 redundant)<br>Hot swappable                                |
| Form factor and dimension | 1 RU<br>(W) 445 mm x (D) 468mm x (H) 44 mm                        |
| Weight, kg                | 9.7 kg                                                            |
| Operating temperature     | 0°C – 40°C                                                        |
| Storage temperature       | -40°C – 70°C                                                      |
| Operating humidity        | 5% – 95% (non-condensing)                                         |
| Storage humidity          | 5% – 95% (non-condensing)                                         |
| Operating altitude        | 3000m max                                                         |
| MTBF                      | 40 years                                                          |
| MTTR                      | 1 hour                                                            |

### Performance and Scalability

|                          |                                                                       |
|--------------------------|-----------------------------------------------------------------------|
| Model                    | S6750-24X6CQM-AC                                                      |
| Switching Capacity       | 2.18 Tbps                                                             |
| Forwarding Rate          | 600Mpps                                                               |
| Jumbo frames             | Yes                                                                   |
| Packet Buffer Size       | 6 MB                                                                  |
| Bridge FDB Entries       | 32K                                                                   |
| Router IPv4 Host Entries | Up to 16K Unicast prefixes (LPM) 1<br>or<br>Up to 3K (*,G) addresses  |
| Router IPv6 Host Entries | Up to 8K Unicast prefixes (LPM) 3<br>or<br>Up to 0.5K (*,G) addresses |

|                                                                           |                                                                                                                                                                                              |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Router LPM                                                                | IPv4 - up to 16K unicast prefixes<br>IPv6 - up to 8K unicast 64-bit prefixes                                                                                                                 |
| TCAM Resources (Shared between TTI, IPCL and EPCL Classification Engines) | 9K x 10B rules                                                                                                                                                                               |
| Virtual Ports (ePorts)                                                    | 8K                                                                                                                                                                                           |
| Bridge Domains (eVLANs)                                                   | 8K                                                                                                                                                                                           |
| Multicast Physical Port Groups (VIDX)                                     | 4K                                                                                                                                                                                           |
| Multicast Virtual Port Groups (eVIDX)                                     | 16K                                                                                                                                                                                          |
| Multicast Linked List (MLL) Entries                                       |                                                                                                                                                                                              |
| Tunnel-Start/ARP Table                                                    | Up to one of the following: <ul style="list-style-type: none"><li>• 32K ARP entries</li><li>• 8K non-IPv6 Tunnel Start table entries</li><li>• 4K IPv6 Tunnel Start table entries.</li></ul> |
| Router Next Hop Entries                                                   | 8K                                                                                                                                                                                           |
| Router ECMP Groups                                                        | 8K                                                                                                                                                                                           |
| ePort ECMP Groups                                                         | 8K                                                                                                                                                                                           |
| Trunk ECMP Group                                                          | 4K                                                                                                                                                                                           |
| Hardware OAM Flows                                                        | 2K                                                                                                                                                                                           |
| Spanning Tree Groups                                                      | 1K                                                                                                                                                                                           |
| QoS Profiles                                                              | 1K                                                                                                                                                                                           |